Convex Prime — the leveraged sibling

Same convex engine, scaled up. What leverage does to the payoff, how it's sized and gated, and the honest trade it makes.

Read the Convex Core tutorial first. Prime is *identical* to Convex Core — the same `w_equity + w_convexity + w_duration = 1` budget, the same ETFs, the same vol-brake and regime engine — with **one** addition: leverage. This page only covers that one difference.

1 · Leverage = amplifying a convex payoff

Naive leverage is dangerous because it amplifies a *linear* exposure — 2× the market means 2× the crash. Prime is different: it levers a book that is already **convex** (the crash hedges are levered too), so it amplifies the *whole* bent payoff of Figure 1 — more upside, and only a little more downside, because the protection scales up with the exposure.

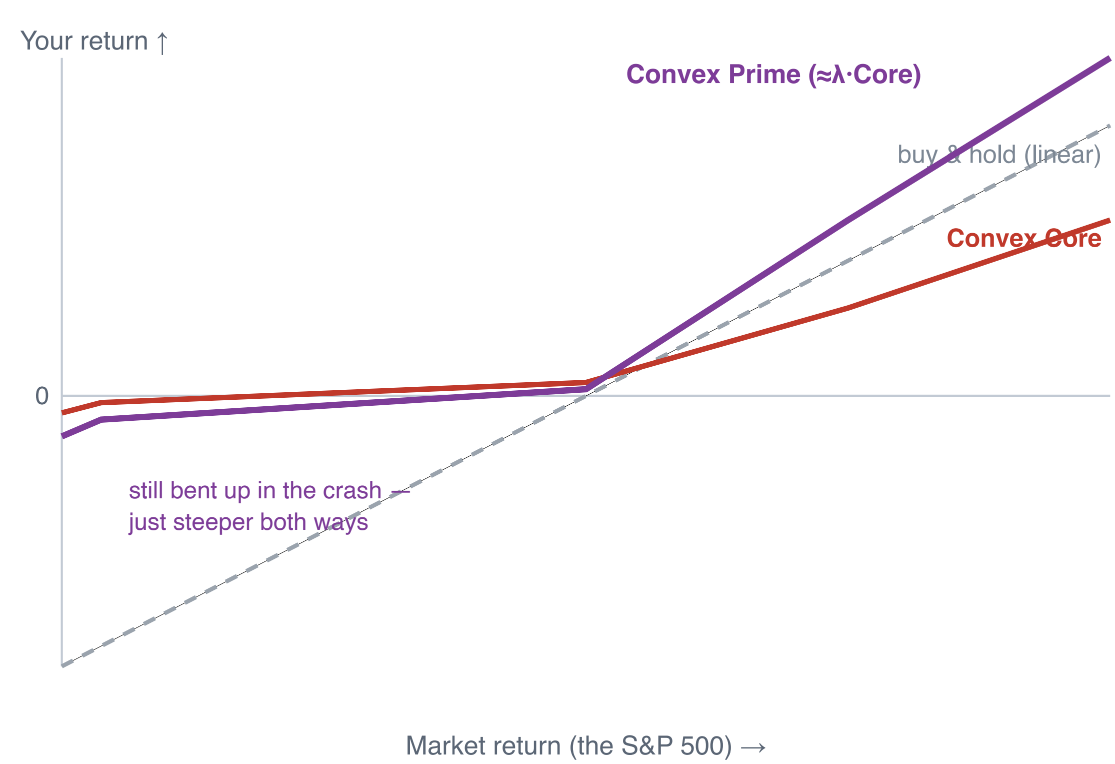

Figure P1. Prime's payoff is roughly **λ × Convex Core's** (here λ≈2). It keeps the convex shape — still far above the straight line in the left tail — but steeper: more upside, a bit more downside. Leverage on a convex book amplifies the *bend*, not just the beta. (Conceptual.)

2 · How the leverage is built — and bounded

Prime doesn't pick a fixed "2×." The multiple is sized and capped to hit a **volatility target**, and a sentiment signal pulls it back when options markets get nervous.

Mechanism	What it does
Financed gross book	Builds exposure above 100% with a financed cash leg — the same sleeves, scaled.
λ sized to a vol target	The leverage multiple **λ** flexes to target ≈**16% annual volatility** (Core targets 12%), and is **hard-capped** at a maximum so it can never run away.
Put/Call-ratio brake	A PCR sentiment signal **cuts exposure when options markets flash caution** — a second, faster throttle on top of the vol-target sizing.
Same convex structure	Identical sleeves + scored tilts + de-dup + incumbency rules as Core (Sections 2–6 of the Core tutorial). Leverage scales the whole book — hedges included.

So the volatility brake (Core's Figure 3) *and* the regime engine (Core's Section 6) still run underneath — leverage sits on top of all of it, sized so the **levered** portfolio targets 16% vol, not the unlevered one.

3 · 20 years of $1 — more destination, rougher road

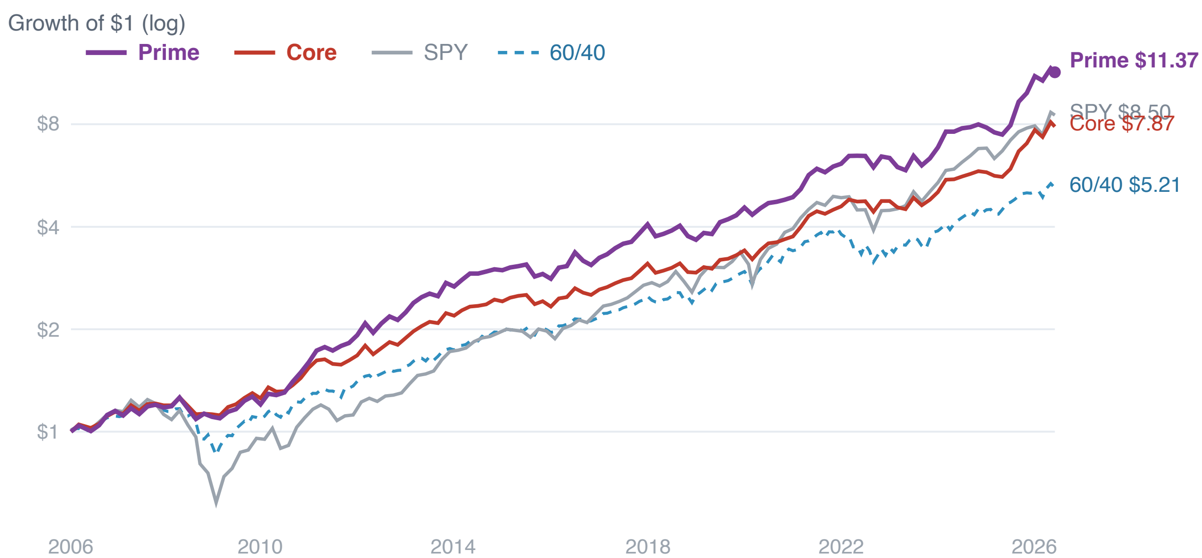

Figure P2. Growth of $1, 2006→2026, **log scale** (published backtest, net of modeled costs). Prime ends highest (**$11.37** vs Core $7.87, SPY $8.50, 60/40 $5.21) — but look at 2008 and 2022: its dips are visibly deeper than Core's. More destination, a rougher road.

4 · The honest trade — it is not a free lunch

Leverage amplifies *everything*, including the small cost the hedges charge in calm markets. So Prime earns **more absolute return** than both Core and SPY, with **much better drawdowns than SPY** — but its **risk-adjusted** efficiency is a touch *below* unlevered Core.

Over ~20 years	SPY	Convex Core	Convex Prime
Growth of $1	$8.50	$7.87	**$11.37**
CAGR	≈11.0%	≈10.7%	**≈12.7%**
Sharpe (risk-adjusted)	≈0.72	**≈1.05**	≈0.95
Worst drawdown	≈–55%	**≈–18%**	≈–21%

Read the trade carefully. Prime beats SPY on return *and* drawdown — because the convex hedges are levered too. But it does **not** beat Core on a risk-adjusted basis: the same leverage that lifts the return also amplifies the hedges' calm-market cost and deepens the drawdowns (–21% vs –18%).

Prime is for an investor who wants more *absolute* return and can genuinely sit through a deeper hole — not a strictly better Core.

5 · Everything else is Convex Core

The equity sleeve (SPY + IC-scored tilts), the convexity sleeve (DBMF/KMLM/BTAL), the regime-switched duration ladder, the volatility brake, the regime engine, and the open question about the equity core (SPY vs VTI/VXUS/IWM, CAPE) are **identical** to Convex Core — leverage just scales the result. For all of that, see the Convex Core tutorial.

> **One-sentence summary:** Convex Prime is Convex Core with a sized, capped, sentiment-gated layer of leverage on top — it amplifies the *bend*, buying more return and keeping crash protection SPY never had, at the cost of deeper drawdowns and slightly lower risk-adjusted efficiency than unlevered Core.

Educational illustration of the DNSR Convex Prime model. Figure P1 is conceptual; the Section-4 statistics, the 0.65/0.15/0.20 structure, the vol-target / PCR-gated leverage, and Figure P2 are from the model engine and the published backtest. **Leverage materially raises risk. Hypothetical, backtested performance — not investment advice; past performance does not guarantee future results.** Convex Prime is a research / observation model, not the IRA implementation. Full method + significance testing in the Convex Core paper (the 📕 / 📄 links on the model card).